Projected Health Impact of Achieving WHO Targets for Cardiovascular Risk-Factor Control

Work in Progress

WHO · University of Washington

September 2026

Study Overview

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Study Overview

Aim 1: Research Questions

Central Question

What is the projected **burden-of-disease impact** of achieving the WHO cardiovascular risk-factor control targets by **2030**?

Sub-questions driving the analysis:

1. How many **cardiovascular events** could be averted?
2. How many **CVD deaths** could be prevented by 2030 and beyond?
3. How would **life-years, YLLs, YLDs, and DALYs** change under joint achievement of:
   - 150M additional people with controlled hypertension
   - $\geq$50% lipid-lowering therapy coverage
   - 80% BP control among people with diabetes
   - 80% validated BP device availability

Scope

186 countries · Ages 20–95 · 2026–2050 · Multiple CVD causes · WHO Target Scenarios

Intervention start: 2026

Model calibrated to 2019 baseline. First year of target scale-up effects entering the projection is **2026**.

Study Design at a Glance

Component	Detail
Model type	Discrete-time multi-state model (Well → Sick → Dead)
Calibration data	GBD 2023: incidence, case fatality, prevalence; all-cause & CVD-specific
Population data	UNWPP 2024 single-year age/sex projections
Intervention start	**2026** — first year intervention effects enter the projection
Scale-up period	Linear scale-up 2026 → 2030; rates held constant 2030–2050
Comparison	Business-as-usual: 2024 control rates, no additional scale-up
Outcome	CVD deaths averted vs. BAU; age-standardised mortality rate (ASMR)
Time horizon	2026–2050 (25-year horizon); model calibrated from 2019

Study Context

Objective

- Quantify CVD deaths avertable by achieving WHO 2030 cardiovascular targets
- Coverage: **186 countries**, 2026–2050
- Four causes: IHD, ischemic stroke, hemorrhagic stroke, hypertensive HD

Model

- Discrete-time state transition (Well $\rightarrow$ Sick $\rightarrow$ Dead)
- Calibrated to GBD 2023; projected with UNWPP 2024 populations
- 8 blood-pressure bins
- Baseline treatment effects explicitly de-duplicated

Research Question

What are the projected deaths averted from achieving WHO hypertension control and lipid-lowering therapy targets across **186 countries**, 2026–2050?

Pending △

Confirm primary metric: deaths averted vs. DALYs vs. LYG

Data Sources

Data Sources and Key Variables

Source	Variables extracted	Role in model
GBD 2023 (IHME)	IR, CF, prevalence, background mortality (age/sex/country)	Baseline transition rates; initial states
GBD 2019 (IHME)	Relative risks per 10 mmHg SBP increase (IHD, stroke, HHD, by age)	Anchor BP-bin incidence; normalisation factor α
Ettehad et al. 2016	Trial effect sizes per BP category; diabetes-stratified	Treatment effect sizes by BP bin $\times$ cause
UNWPP 2024 (UN)	Population projections 2019–2050 (single-year age, sex, country)	Cohort sizes; rate denominators
NCD-RisC	Baseline HTN control rates (C_0); BP distribution by country	C_0; $P(\text{BP}_{\text{cat}})$
Country-specific	HTN control targets 2030 (Progress / Ambitious / Aspirational)	C_{target} per scenario

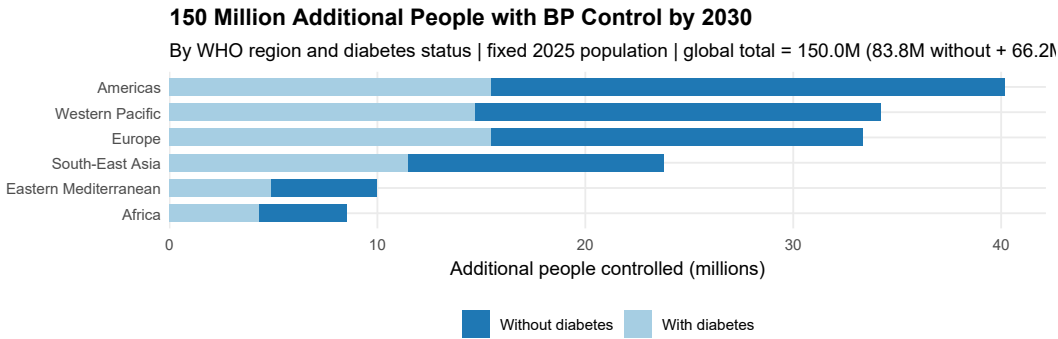
Scenarios

Four WHO Priority Targets

#	WHO Target	Modelled Scenario
1	150M more with HTN controlled by 2030	**HTN Control (Combined)**: 150.0M more controlled = 83.8M without + 66.2M with diabetes (mutually exclusive, additive)
2	$\geq$50% eligible adults on lipid-lowering therapy by 2030	**Improved Statin Uptake**: 50% coverage by 2030
3	80% BP control among diagnosed diabetics by 2030	**HTN Control (Diabetes)**: 0.80 aspirational ceiling by 2030 (66.2M of the 150M)
4	80% availability of validated BP devices	Operationally embedded in HTN scenarios (150M Target [1+3])

Combination rule: the two BP subgroups combine *additively* as a population mixture, $\varepsilon_{IR,\text{BP}} = 1 - (1 - w_D)e_N - w_D e_D$; the BP composite then combines multiplicatively with statins, $\varepsilon_{IR,\text{total}} = \varepsilon_{IR,\text{BP}} \times \varepsilon_{IR,\text{statin}}$.

The 150 Million Target — Additional BP Control by Region and Diabetes Status



HTN Control Target: 150 Million by WHO Region

Region	HTN pop 2025 (M)	Controlled 2025 (M)	Cov. 2025	Controlled 2030 (M)	Cov. 2030	Added (M)	+%
Africa	182.0	20.6	11.3%	29.1	16.0%	8.5	41.5%
Americas	273.9	100.4	36.7%	140.5	51.3%	40.1	40.0%
Eastern Mediterranean	157.5	24.0	15.2%	34.0	21.6%	10.0	41.5%
Europe	300.3	80.5	26.8%	113.9	37.9%	33.3	41.4%
South-East Asia	425.8	57.3	13.4%	81.1	19.0%	23.8	41.5%
Western Pacific	456.1	83.5	18.3%	117.7	25.8%	34.2	41.0%
Global	**1,795.5**	**366.2**	**20.4%**	**516.2**	**28.8%**	**150.0**	**41.0%**

Coverage rates and 2030 counts use the fixed 2025 hypertensive population; regional additions sum to exactly 150 million.

HTN Control Target: 150 Million by WHO Region *and* Diabetes Subgroup

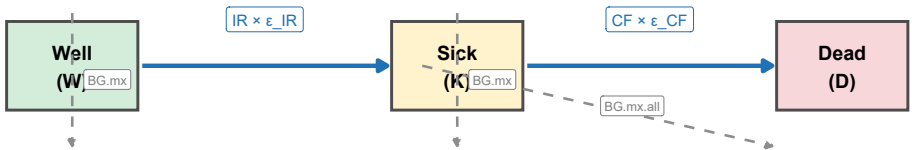
	Additional controlled 2030 (M)				
Region	**No diabetes**	**Diabetes**	**Total**	**Cov. 2030**	**+%**
Africa	4.3	4.3	8.5	16.0%	41.5%
Americas	24.7	15.5	40.1	51.3%	40.0%
Eastern Mediterranean	5.1	4.9	10.0	21.6%	41.5%
Europe	17.9	15.4	33.3	37.9%	41.4%
South-East Asia	12.3	11.5	23.8	19.0%	41.5%
Western Pacific	19.5	14.7	34.2	25.8%	41.0%
Global	**83.8**	**66.2**	**150.0**	**28.8%**	**41.0%**

Additional people with BP control by 2030, disaggregated by WHO region *and* mutually exclusive diabetes status (fixed 2025 population). Both the regional totals and the two subgroup totals sum to exactly 150 million.

Mathematical Model

Model Architecture: Three Health States

Multi-State Transition Model | Annual time step | Ages 20-95 | 4 CVD causes



Key: $\varepsilon_{IR}, \varepsilon_{CF}$ = multiplicative effect ratios from the intervention module · BG.mx = background mortality · COVID-19 excess mortality applied 2020–2021 only

Modeling Assumptions

Model structure

- Discrete-time annual step (2019–2050); ages 20–95
- Four causes modeled independently
- COVID-19 excess mortality applied 2020–2021 only; no long-term effects

Intervention mechanics

- Linear scale-up 2026 $\rightarrow$ 2030; held constant 2030–2050
- Only incremental coverage above C_0 (or $U_{0,\ell}$) generates new effects (no double-counting)
- Full adherence assumed for HTN control, statins adherence adjusted down for primary and secondary prevention.
- Statin effects restricted to ages ≥ 40 and IHD & ischemic stroke only

Blood pressure & effect sizes

- Static log-normal BP distribution: no secular trend, no endogenous updating as coverage rises
- Treatment eligibility: $\geq$140 mmHg (5 hypertensive bins)
- GBD RRs anchor bin-level baseline incidence
- Hypertensives split into mutually exclusive no-diabetes and diabetes subgroups; a person with diabetes is counted only in the diabetes subgroup
- Baseline control assumed 15 percentage points higher with diabetes, preserving the observed overall rate: $c_N = c - w_D(0.15)$, $c_D = c + (1 - w_D)(0.15)$
- Subgroup-specific Ettehad effect sizes applied to each subgroup's own share, then combined *additively* as a population mixture (no cross-term)

Combination of interventions

- The two BP subgroups combine **additively**:

Burden of Disease: Methods

Three burden-of-disease metrics

- **YLL** (Years of Life Lost): deaths $\times$ remaining life expectancy at age of death
- **YLD** (Years Lived with Disability): prevalent CVD cases $\times$ disability weight
- **DALY** (Disability-Adjusted Life Years): $DALY = YLL + YLD$

DALYs averted = $DALY_{\text{BAU}} - DALY_{\text{intervention}}$

Full derivation in Supplemental Mathematical Details.

Reference standards

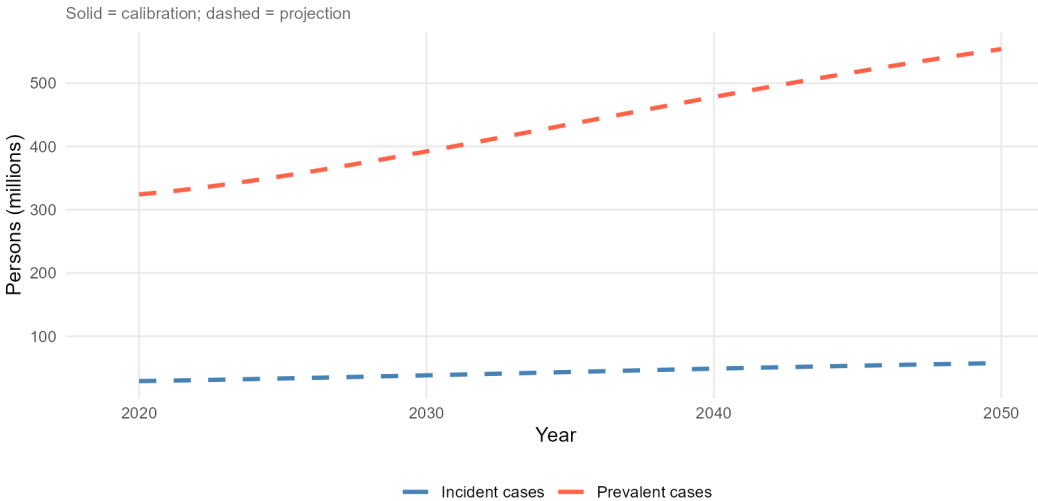
Life table: UNWPP 2024, medium variant (country- and year-specific e_x)

Disability weights: GBD 2023, derived as YLD/Prevalence per cause and location

Discounting: None (undiscounted)
Age weighting: None

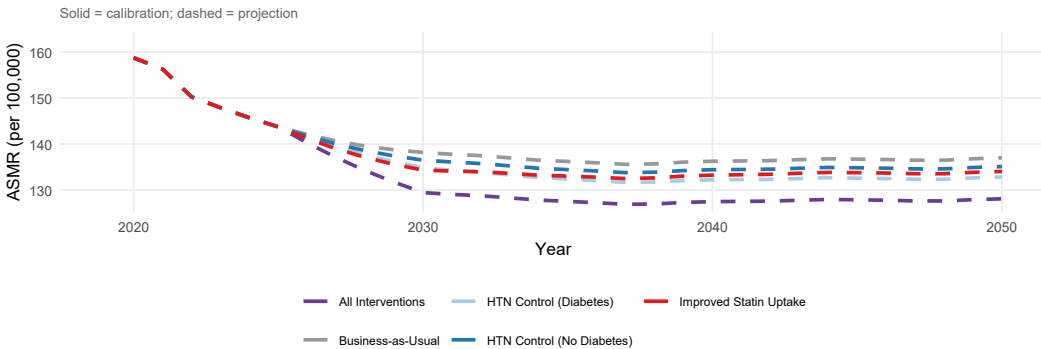
Results

Baseline Disease Trends (Business-as-Usual)



Incident and prevalent CVD cases 2019–2050; solid = calibration, dashed = projection.

Age-Standardised CVD Mortality Rate



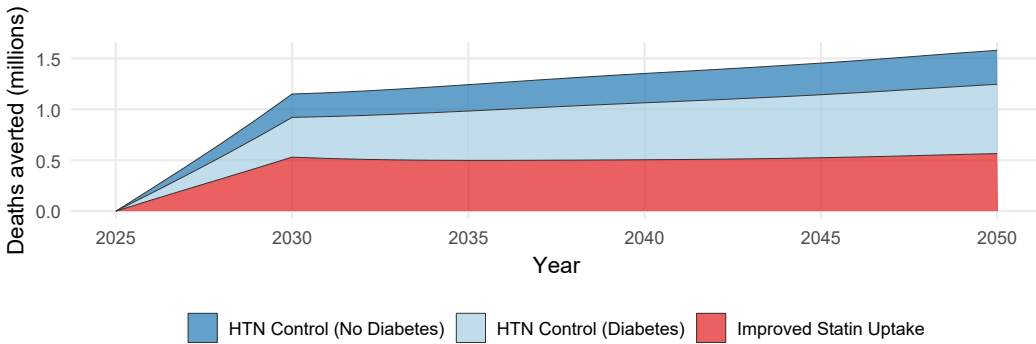
Summary: Cumulative Deaths Averted, 2026–2050

Scenario	Cumulative (M)	Avg./year (M)	% of BAU deaths
HTN Control (No Diabetes)	6.4 M	0.3 M	1%
HTN Control (Diabetes)	12.3 M	0.5 M	1.9%
Improved Statin Uptake	11.9 M	0.5 M	1.9%
All Interventions	**30.5 M**	**1.2 M**	**4.8%**

Headline

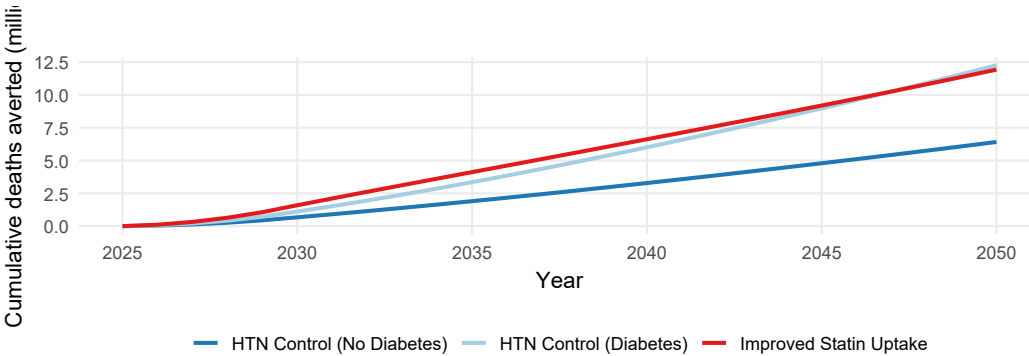
All Interventions: **30.5 million deaths averted** globally, 2026–2050 ($\approx$ 1.2 M/year).

Annual Deaths Averted Over Time



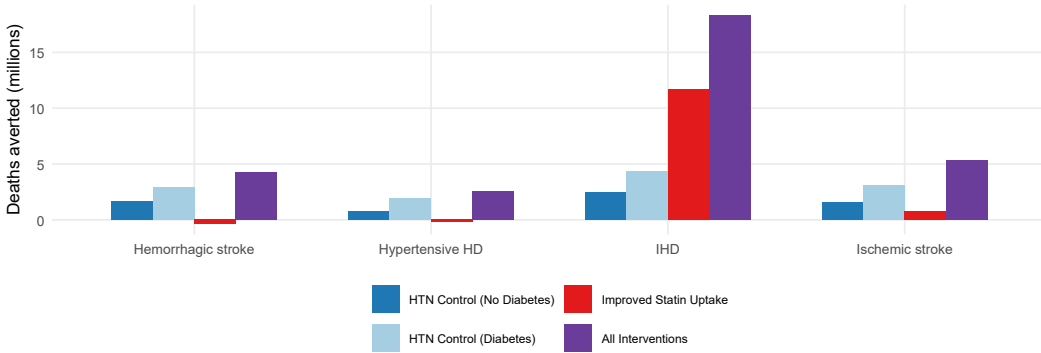
Cumulative deaths averted per intervention relative to business-as-usual.

Cumulative Deaths Averted Over Time

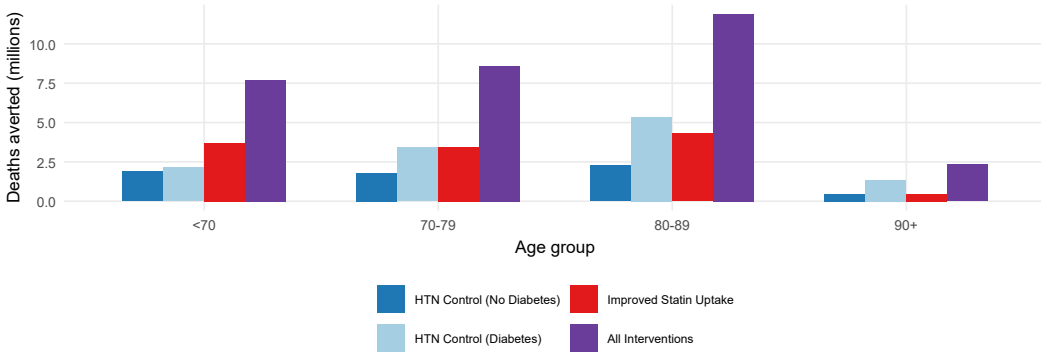


Cumulative deaths averted per intervention relative to business-as-usual.

Deaths Averted by Cause of Death



Deaths Averted by Age Group



Burden of Disease Averted: YLLs, YLDs, and DALYs

- DALYs averted: 360.8 M
- YLLs averted: 356.6 M
- YLDs averted: 4.2 M

Intervention	Deaths averted	DALYs averted	YLDs averted	YLLs averted
HTN Control (No Diabetes)	6.4 M	84.1 M	1.8 M	85.9 M
HTN Control (Diabetes)	12.3 M	123.7 M	3.8 M	127.5 M
Improved Statin Uptake	11.9 M	151.2 M	-1.4 M	149.8 M
All Interventions	30.5 M	356.6 M	4.2 M	360.8 M

Headline

The combined WHO target package generates large health gains, with most benefits coming through avoided premature CVD deaths rather than morbidity reduction alone.

Regional Summary

Region	BAU deaths (M)	Averted (M)	% averted
Western Pacific	240.3	10.5	4.4%
South-East Asia	139.2	5.7	4.1%
Europe	103.7	5.5	5.3%
Americas	58.9	4.4	7.4%
Eastern Mediterranean	53.3	2.6	4.8%
Africa	41.7	1.8	4.3%

Observation

Absolute gains largest in high-burden regions; proportional gains reflect current coverage gaps.

Economic Value of Avoidable Mortality

Economic Valuation: Methods at a Glance

Two complementary metrics

- **Value of Statistical Life (VSL)** — monetary value the population places on small reductions in mortality risk; transferred from a U.S. reference using country GNI per capita (PPP) and an income elasticity.
- **Value of Statistical Life Year (VSLY)** — VSL annuitised by remaining life expectancy at the average adult age, then applied to age-specific life-years gained from deaths averted.

Headline framing: VSLY is reported as a **proportion of annual income** (discounted GNI) so results are directly comparable across regions and over time.

Reference-case parameters

Income elasticity (primary): differential 0.8 (HIC) / 1.2 (LMIC), per Robinson et al. (2019)
Sensitivity: uniform $e = 1.5$
Discount rate: 3% (calendar-year)
Income projection: World Bank GNI (PPP), forward-projected with SSP2 GDP growth
Horizon: 2026–2050

Methodological alignment

Approach follows **Robinson et al. (2019)** guidance and is consistent with recent published modelling of CVD policy benefits in World Bank's Healthy Longevity Initiative *Nature Medicine* papers (Chang et al. 2024; Verguet et al. 2024).

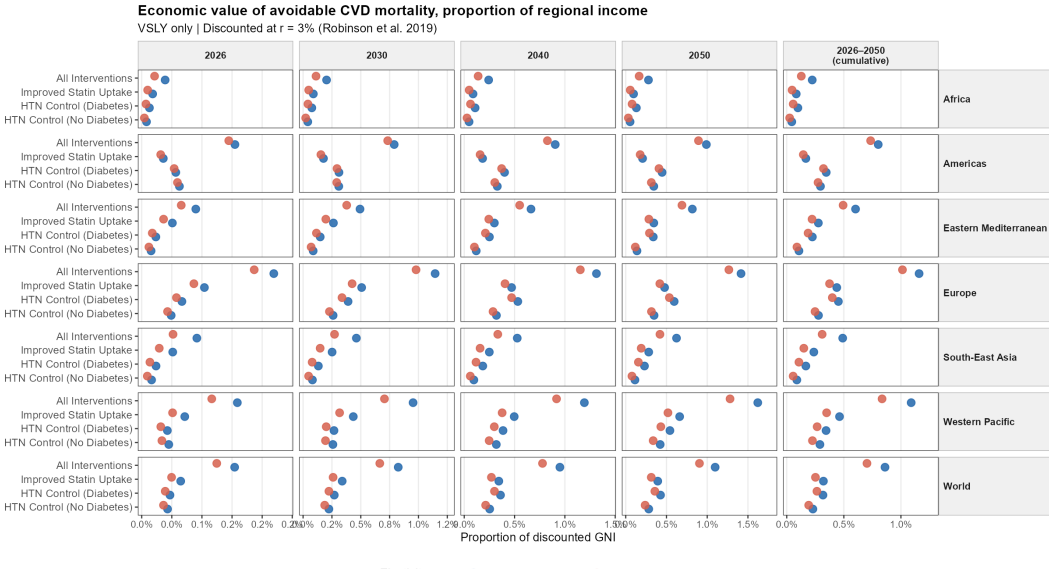
Economic Value by WHO Region (Primary, Cumulative 2026–2050)

World: All Interventions, 2026–2050 (primary, discounted at 3%)

VSLY = **0.9%** of annual income PPP (cumulative **$35.3 T**).

Region	VSLY (USD T)	VSLY % GNI
Africa	$0.5	0.2%
Americas	$7.2	0.8%
Eastern Mediterranean	$1.8	0.6%
Europe	$10.5	1.2%
South-East Asia	$3.4	0.5%
Western Pacific	$12.0	1.1%
World	**$35.3**	**0.9%**

Value of Avoidable Mortality as a Proportion of Income



Discussion

Interpretation

Key findings

- **All Interventions** yields largest gains: HTN control + statins are complementary
- **IHD** and **ischemic stroke** drive the majority of avertable deaths
- Gains **accelerate after 2030** as coverage compounds
- **Older adults (70+)** benefit most in absolute terms
- **Top 15 countries** account for >75% of avertable burden

Policy Implication

Combined WHO Targets 1 + 2 are synergistic — neither alone captures the full benefit.

Uncertainty △

Sensitivity analysis to diabetes targeting assumptions and secular trends pending.

Limitations

Model structure & assumptions

- Stable transition rates assumed; no dynamic feedback between states
- Static BP distribution (log-normal): no secular trend and no endogenous updating as coverage increases
- No heterogeneity in adherence: newly HTN covered individuals assumed fully adherent
- COVID-19 excess mortality only modelled for 2020–2021; long-term effects excluded

Data & parameters

- RCT-based effect sizes (e.g., Ettehad ρ_{CF}): real-world effectiveness at scale may be lower
- GBD 2023 uncertainty intervals not propagated
- Baseline C_0 from NCD-RisC; measurement error unquantified
- BP distributions rely on GBD modelling (not direct surveys in all countries)

Scenarios & implementation realism

- Linear scale-up assumed (2026–2030); real programmes may follow non-linear trajectories
- No explicit health-system capacity constraints modelled
- Interactions with statins, diabetes care, and BP device availability not included

Next Steps

Analysis priorities

1. Sensitivity analysis on effect sizes, trends and coverage
2. Add life-years gained (LYG) and DALYs as co-primary outcome
3. Stratify by income group (World Bank)
4. Conduct Aim 2: simulate alternative scale-up trajectories and implementation scenarios for hypertension control

Pending Decisions △

- Journal target?

Summary

1. The health gains are large and achievable

All Interventions: **30.5 million deaths averted** globally, 2026–2050.

2. HTN control + statins compound risk reduction

HTN control and statins have multiplicative risk reductions, producing larger combined effects.

3. Regional patterns reflect burden and coverage gaps

Absolute gains largest in high-burden regions; proportional gains reflect current coverage gaps.

4. The economic value rivals annual income itself

World, All Interventions, 2026–2050: avoidable CVD mortality is worth **0.9% of annual income** (VSLY, cumulative **$35.3 T**, discounted at 3%).

Meeting WHO 2030 CVD targets is not just aspirational — it is quantifiably lifesaving and economically transformative

Appendix

Notation: Core Model

Symbol	Description
Core state-transition model	
a, s, c, t, ℓ	Age, sex, cause, year, location
W, K, D	Well, Sick, Dead populations
$IR_{a,s,c}$, $CF_{a,s,c}$	Baseline incidence & case-fatality rates
$\varepsilon_{IR,j}$, $\varepsilon_{CF,j}$	Multiplicative effect ratios for intervention j
$BG.mx$, $BG.mx.all$	Cause-specific & all-cause background mortality
$covid.mx$	COVID-19 mortality (2020–2021 only)

Notation: HTN Control

Symbol	Description
HTN control (general & diabetes subgroup)	
C_0, C_t, C_{target}	Baseline, time-t, and target control rates
ΔC_{target}, $\Delta C_{\text{increment}}$	Total & time-varying incremental coverage
C^{Δ}_{agg}	Population-weighted incremental coverage (hypertensive bins)
$P(\text{BP}_{\text{cat}})$	BP category population distribution (8 bins)
RR_{GBD}	GBD relative risk per 10 mmHg SBP increase
α_{inc}	Incidence normalization factor
E_D, E_N	Ettehad effect sizes (diabetes, no-diabetes subgroups)
$e_{g,t}$	Baseline-adjusted incremental effect for subgroup $g \in \{N, D\}$
w_D	Diabetes share among hypertensives ($= P(D \mid H)$, mixture weight)
c_N, c_D	Baseline control without / with diabetes ($c_D - c_N = 0.15$)
$\rho_{CF,c}$	CF reduction factor per unit incremental coverage
T_{start}, T_{end}	Scale-up start (2026) & end (2030)

Notation: Statins & Combined Effects

Symbol	Description
Statin therapy	
$U_{0,\ell}$, U_{target}	Baseline & target statin coverage
$\Delta U(t,\ell)$	Incremental statin coverage above $U_{0,\ell}$ at time t
$T_{\text{stat,start}}$, $T_{\text{stat,end}}$	Statin scale-up start (2026) & end years
RR_c^{IR}, RR_c^{CF}	Trial-based RRs for incidence & case fatality
δ_c^{IR}, δ_c^{CF}	Risk reductions $= 1 - RR$
$AF_{c,\ell}$	Cause- & location-specific attributable fraction (incidence only)
ψ_{athero}	Atherosclerotic fraction of ischemic stroke ($= 0.60$)
$E_{c,\ell}^{IR}(t)$, $E_{c,\ell}^{CF}(t)$	Incremental effect sizes for IR and CF

HTN Intervention Module: 9-Step Pipeline

	Action	Output
1	Baseline BP distribution $P(\text{BP}_{\text{cat}})$ without treatment (rx $= 0$)	$P(BP_{cat} \mid a, s, t)$
2	GBD relative risks per 10 mmHg SBP $\rightarrow$ bin-specific incidence IR_{bin}	IR_{bin}
3	Subgroup-specific Ettehad effect sizes E_N, E_D (no-diabetes, diabetes) per BP bin $\times$ cause	E_{trial}
4	Linear coverage scale-up from C_0 to C_{target}, **starting 2026**	$C_t(t,\ BP_{cat})$
5	Coverage adjustment: $E_{\text{adj}} = E_{\text{trial}}(C_t - C_0) \,/\, (1 - E_{\text{trial}}\, C_0)$	$E_{adj}(t)$
6	New bin incidence: $IR_{\text{bin,new}} = IR_{\text{bin}} \times (1 - E_{\text{adj}})$	$IR_{bin,new}$
7	Aggregate: $IR_{\text{new}} = \sum IR_{\text{bin,new}} \times P(\text{BP}_{\text{cat}})$	IR_{new}
8	Incidence effect ratio: $\varepsilon_{IR} = IR_{\text{new}} \,/\, IR_{\text{original}}$	$\varepsilon_{IR} \rightarrow$ *model*
9	Case-fatality effect ratio: $\varepsilon_{CF} = 1 - \rho_{CF} \times C_{\text{agg}}^{\Delta}$	$\varepsilon_{CF} \rightarrow$ *model*

Blue row = intervention start (2026). Green rows = outputs passed to state-transition model.

HTN Control: Coverage Scale-Up and Key Equations

Incidence pathway

Step 2 — GBD RR anchors bin incidence:

$$IR_{\text{bin}} = \frac{RR_{\text{GBD}} \times IR}{\alpha}, \quad \alpha = \sum_{\text{cat}} P_{\text{cat}} \cdot RR_{\text{GBD}}$$

Step 5 — baseline-adjusted effect (no double-counting):

$$E_{\text{adj}}(t) = \frac{E_{\text{trial}}\,(C_t - C_0)}{1 - E_{\text{trial}}\,C_0}$$

Steps 7–8 — population-weighted effect ratio:

$$\varepsilon_{IR} = \frac{\sum IR_{\text{bin,new}} \cdot P_{\text{cat}}}{IR_{\text{original}}}$$

Case fatality pathway

Incremental coverage above baseline:

$$\Delta C_\delta(t) = \max(C_t - C_0,\ 0)$$

Aggregate across hypertensive bins (≥140 mmHg):

$$C^{\Delta}_{\text{agg}} = \frac{\sum_{\geq 140} \Delta C_\delta \cdot P_{\text{cat}}}{\sum_{\geq 140} P_{\text{cat}}}$$

CF effect ratio:

$$\varepsilon_{CF} = 1 - \rho_{CF} \times C^{\Delta}_{\text{agg}}$$

Cause	ρ_{CF}
IHD	0.24
Ischemic stroke	0.36
Hemorrhagic stroke	0.76
HHD	0.20

HTN Control: Mutually Exclusive Diabetes / No-Diabetes Subgroups

Two mutually exclusive, exhaustive subgroups of the hypertensive population; each uses its own Ettehad effect size applied *only* to its own share.

Subgroup-specific incremental effect ($g \in \{N, D\}$):

$$e_{g,t} = \frac{E_g\,(c_{g,t} - c_{g,0})}{1 - E_g\,c_{g,0}}$$

Baseline decomposition (preserves observed rate c):

$$c_N = c - w_D(0.15), \qquad c_D = c + (1 - w_D)(0.15)$$

Diabetes coverage scale-up toward the 0.80 ceiling:

$$c_{D,t} = c_{D,0} + \Delta c_D \;\; \min\left[\tfrac{t-2026}{4},\, 1\right]$$

Additive population mixture (w_D = diabetes share among hypertensives), *one* composite BP effect — no cross-term:

$$\varepsilon_{IR,\text{BP}} = 1 - (1 - w_D)\,e_{N,t} - w_D\,e_{D,t}$$

$$\Delta c_t = (1 - w_D)\,\Delta c_{N,t} + w_D\,\Delta c_{D,t}$$

Why mutually exclusive?

People with diabetes have higher baseline control and larger DM-specific Ettehad effects. Counting each hypertensive once (diabetes *or* not) avoids double-counting and lets the two subgroup additions sum to exactly 150 million.

Combined with statins

The BP composite then combines multiplicatively with the (independent) statin effect:

Statin Therapy: Coverage Scale-Up and Key Equations

Scope: IHD & ischemic stroke; ages ≥ 40 only

Incremental coverage above baseline $U_{0,\ell}$:

$$\Delta U(t,\ell) = \Delta U_{\text{target},\ell} \ \min\left[\frac{t - T_{\text{stat,start}} + 1}{T_{\text{stat,end}} - T_{\text{stat,start}} + 1}, 1\right]$$

where $\Delta U_{\text{target},\ell} = \max(U_{\text{target}} - U_{0,\ell},\ 0)$

Trial-based RRs (per ≈1 mmol/L LDL-C reduction):

Cause	RR^{IR}	RR^{CF}
IHD	0.74	0.80
Ischemic stroke	0.80	0.96

Incidence (primary prevention, AF-scaled):

$$E^{IR}_{c,\ell}(t) = \frac{AF_{c,\ell}\ \delta^{IR}_c\ \Delta U(t,\ell)}{1 - \delta^{IR}_c\ U_{0,\ell}}$$

Case fatality (secondary prevention; no AF; atherosclerotic restriction for stroke):

$$E^{CF}_{c,\ell}(t) = \begin{cases} \dfrac{\delta^{CF}_{\text{ihd}}\ \Delta U(t,\ell)}{1 - \delta^{CF}_{\text{ihd}}\ U_{0,\ell}}, & c = \text{ihd} \\ \dfrac{\psi_{\text{athero}}\ \delta^{CF}_{\text{ist}}\ \Delta U(t,\ell)}{1 - \delta^{CF}_{\text{ist}}\ U_{0,\ell}}, & c = \text{istroke} \end{cases}$$

$\delta^{IR}_c = 1 - RR^{IR}_c$; $\psi_{\text{athero}} = 0.60$
(atherosclerotic fraction)

Effect ratios passed to model:

$$\varepsilon_{IR,\text{stat}} = \begin{cases} 1 - E^{IR}_{c,\ell}(t), & a \geq 40,\ c \in \{\text{ihd,ist} \\ 1, & \text{otherwise} \end{cases}$$

$$\varepsilon_{CF,\text{stat}} = \begin{cases} 1 - E^{CF}_{c,\ell}(t), & a \geq 40,\ c \in \{\text{ihd,is} \\ 1, & \text{otherwise} \end{cases}$$

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Thank You

Questions & Discussion

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